

# Proliferative Radiation Retinopathy after Plaque Radiotherapy for Uveal Melanoma

Carlos Bianciotto, MD, Carol L. Shields, MD, Cesare Pirondini, MD, Arman Mashayekhi, MD, Minoru Furuta, MD, Jerry A. Shields, MD

**Purpose:** To determine risk factors, occurrence rate, management, and outcome of proliferative radiation retinopathy (PRR) after plaque radiotherapy for uveal melanoma.

**Design:** Case-control study.

**Participants:** Three thousand eight hundred forty-one patients who underwent plaque radiotherapy for uveal melanoma were entered into the study.

**Methods:** Retrospective review of medical records.

**Main Outcome Measures:** Proliferative radiation retinopathy after plaque radiotherapy for uveal melanoma.

**Results:** Of 3841 eyes treated with plaque radiotherapy for uveal melanoma, PRR developed in 5.8% at 5 years and in 7% at 10 and 15 years using Kaplan-Meier analysis. The mean time to onset of PRR was 32 months (median, 30 months; range, 4–88 months). On univariate analysis, baseline factors predictive of PRR ( $P < 0.05$ ) included young age, diabetes, hypertension, Hispanic race, shorter tumor distance to the optic disc and to the foveola, Bruch's membrane rupture, choroidal location of the tumor, subretinal fluid, higher radiation dose to the optic nerve and to the foveola, higher radiation rate to the tumor apex and to the tumor base, additional transpupillary thermotherapy, and notched plaque. In the multivariate model, young age (odds ratio [OR], 1.44; 95% confidence interval [CI], 1.25–1.67, per decade decrease), diabetes mellitus (OR, 2.73; 95% CI, 1.69–4.40), and shorter tumor distance to the optic disc (OR, 1.10; 95% CI, 1.04–1.17) were related to the occurrence of PRR. The most common forms of management included panretinal photocoagulation (70%), vitrectomy (21%), and observation (17%). Resolution of the neovascularization was obtained in 63% of eyes after treatment.

**Conclusions:** Proliferative radiation retinopathy developed in 7% of eyes by 10 years after plaque radiotherapy for uveal melanoma. The main factors for development of PRR included young age, preexistent diabetes mellitus, and shorter tumor distance to the optic disc.

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Retinal neovascularization is the formation of new vessels in response to the release of proangiogenic factors.<sup>1–4</sup> It can be found in disorders that cause an alteration in retinal vasculature such as diabetes mellitus, retinal vascular occlusion, and retinopathy of prematurity or in association with intraocular neoplasms such as medulloepithelioma.<sup>3</sup> If untreated, retinal neovascularization can lead to profound vision loss through vitreous hemorrhage or tractional retinal detachment.<sup>5,6</sup>

Although uncommon, retinal neovascularization can develop after ocular irradiation, a condition referred to as proliferative radiation retinopathy (PRR).<sup>7–13</sup> It has been described after external beam radiation therapy, proton beam radiotherapy, helium ion radiotherapy, gamma knife radiotherapy, and plaque brachytherapy.<sup>8–15</sup> In the ophthalmic literature, there are several reports of nonproliferative radiation retinopathy,<sup>9–14</sup> but specific case series with evaluation of risk factors for PRR are uncommon.<sup>7,8</sup> In a recent report of 650 patients with medium choroidal melanomas treated with iodine 125 (<sup>125</sup>I) plaque brachytherapy in the Collaborative Ocular Melanoma Study, Boldt et al<sup>7</sup> found neovascularization at the disc (NVD) in 5.2% of patients at 5 years. Neovascularization elsewhere in the retina (NVE)

was not studied and small and large melanomas were not included in that study.<sup>7</sup>

This article reviews the authors' 28-year experience with PRR after plaque radiotherapy for all uveal melanoma cases from a single institution to identify the occurrence rate, risk factors, management, and outcome of this condition. In this all-inclusive analysis, uveal melanoma of any size and location was included for a comprehensive perspective.

## Patients and Methods

The medical records of patients with uveal melanoma who were treated with plaque radiotherapy between June 1, 1978, and April 30, 2006, at the Ocular Oncology Service of Wills Eye Institute, Thomas Jefferson University, Philadelphia, Pennsylvania, were reviewed. Those patients with the diagnosis of PRR were selected for further analysis. The primary outcome of this analysis was the development of PRR, which was defined as the presence of clinically visible or fluorescein angiographically demonstrated NVD or NVE.

Baseline patient data included age, race, gender, history of diabetes mellitus, systemic hypertension, and hypercholesterolemia. Ocular findings recorded at the initial examination before brachytherapy treatment included best-corrected Snellen visual acuity, previous treatment of the tumor before plaque radiotherapy,

Table 1. Proliferative Radiation Retinopathy after Plaque Radiotherapy for Posterior Uveal Melanoma: Patient Demographics

| Features             | Proliferative Radiation Retinopathy Group (n = 138), No. (%) | Control Group (n = 3703), No. (%) | P Value* |
|----------------------|--------------------------------------------------------------|-----------------------------------|----------|
| Age (yrs)            |                                                              |                                   |          |
| Median (mean, range) | 53 (52, 12–86)                                               | 59 (59, 13–98)                    | <0.001   |
| Race                 |                                                              |                                   |          |
| White                | 132 (96)                                                     | 3634 (98)                         |          |
| Black                | 2 (1)                                                        | 20 (<1)                           | 0.657    |
| Hispanic             | 4 (3)                                                        | 36 (<1)                           | 0.001    |
| Asian                | 0 (0)                                                        | 13 (<1)                           | 0.955    |
| Gender               |                                                              |                                   | 0.411    |
| Male                 | 72 (52)                                                      | 1872 (51)                         |          |
| Female               | 66 (48)                                                      | 1831 (49)                         |          |
| Medical history†     |                                                              |                                   |          |
| Diabetes             | 28 (20)                                                      | 328 (9)                           | <0.001   |
| Hypertension         | 38 (28)                                                      | 845 (23)                          | 0.005    |
| Hypercholesterolemia | 9 (7)                                                        | 204 (6)                           | 0.079    |

\*Cox Regression analysis.

†Some patients had more than one condition.

and the presence and type (proliferative vs. nonproliferative) of diabetic retinopathy.

The collected tumor data included tumor location (choroid, ciliary body, iris), quadrant of tumor epicenter (superior, nasal, inferior, temporal, or macular), distance (millimeters) of the posterior margin of the tumor to the optic disc and to the foveola, maximum tumor diameter (based on indirect ophthalmoscopy and B-scan ultrasonography), initial thickness (based on A- and B-scan ultrasonography), tumor shape (dome, mushroom, or diffuse), color (pigmented, partially pigmented, and nonpigmented), associated features (Bruch's membrane rupture, retinal invasion, and extraocular extension of the tumor), and the presence of subretinal fluid.

Plaque radiotherapy data included radioactive isotope ( $^{125}\text{I}$ , ruthenium 106, cobalt 60, and iridium 192 [ $^{192}\text{Ir}$ ]), plaque shape (round or notched), plaque size, hours of radiation, and radiation dose and radiation rate to the tumor apex, tumor base, optic disc, foveola, and lens. Associated treatment with transpupillary thermotherapy (TTT) was recorded.

The time to onset of PRR was recorded. The follow-up examinations were at 4-month intervals for 2 years, then at 4- to 6-month intervals until 5 years, and then at 6- to 12-month intervals thereafter. The development of associated vitreous hemorrhage (VH), diabetic retinopathy, and retinal vascular occlusion was recorded. The number of treatments and treatment methods used for management of PRR were recorded (observation, panretinal photocoagulation [PRP], cryotherapy, intravitreal triamcinolone, and pars plana vitrectomy). The PRP laser was applied to the areas of retinal ischemia. At date last seen, the visual acuity and final outcome for PRR were recorded (PRR was graded as completely resolved, partially resolved, stable, or progressive). Other radiation-associated side effects recorded included nonproliferative retinopathy, maculopathy, papillopathy, scleral necrosis, glaucoma, and cataract.

## Statistical Analysis

The factors associated with PRR were analyzed using Cox regression analysis. The hazard ratios were estimated for the associated variables using Cox regression analysis. First, each variable was investigated in a univariate manner. All variables that were significant on a univariate level ( $P < 0.05$ ) were included in a multivariate model. Kaplan-Meier survival analysis was performed to estimate the survival probabilities of the development of PRR.

## Results

Between June 1978 and April 2006, there were 3841 patients (3841 eyes) with uveal melanoma who underwent plaque radiotherapy at the Oncology Service, Wills Eye Institute. Of the entire group of 3841 patients, PRR developed in 138 (138 eyes; 3.6%) during subsequent follow-up and were included in the PRR group. In the remaining 3703 patients, PRR did not develop, and these patients were included in the no-PRR group (Tables 1–4). Proliferative radiation retinopathy was in the form of NVE in 70 patients (51%), NVD in 46 patients (33%; Fig 1), and a combination of both in 22 patients (16%). The median interval between the application of the plaque radiotherapy and the diagnosis of PRR was 30 months (range, 4–88 months).

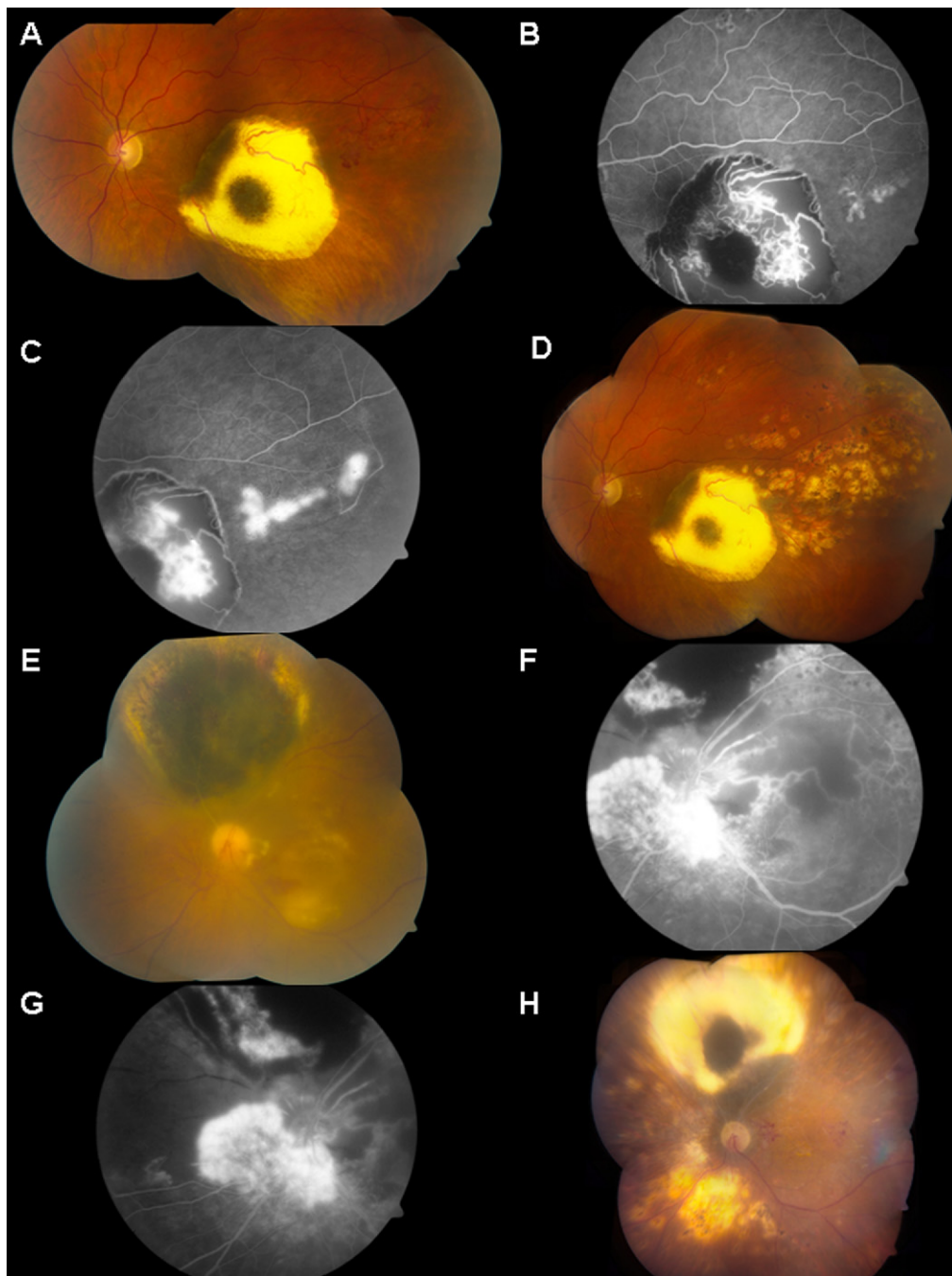
Ocular findings in the PRR patients showed an initial visual acuity before brachytherapy of 20/20 to 20/40 in 89 patients (65%), 20/50 to 20/100 in 32 patients (23%), and 20/200 or worse in 17 patients (12%). Four patients (3%) had undergone previous tumor treatment with TTT and 11 patients (8%) previously had been observed. Five patients (4%) had a previous history of nonproliferative diabetic retinopathy at the first visit.

The demographic features are listed in Table 1. The median patient age at treatment in the PRR group was 53 years versus 59 years in no-PRR group. Diabetes mellitus was present in 20% of patients with PRR and in 9% of patients in the no-PRR group.

Tumor features are listed in Table 2. Choroidal location of the tumor was found in 95% of the patients in the PRR group versus 85% of patients in the no-PRR group. Median tumor distance to the optic disc was 1.9 mm in the PRR group versus 4 mm in the no-PRR group, whereas median tumor distance to the foveola was 2.5 mm in the PRR group versus 3.5 in the no-PRR group. A break in Bruch's membrane was found in 29% of patients in the PRR group versus 15% in the no-PRR group.

Plaque radiotherapy parameters are listed in Table 3. A notched plaque shape was used in 43% of patients in the PRR group versus 23% in the no-PRR group. Mean radiation dose to the optic disc was 7164.21 cGy in the PRR group versus 5016.45 cGy in the no-PRR group. Mean radiation dose to the foveola was 9568.46 cGy in the PRR group versus 6676.99 cGy in the no-PRR group.

Of the 138 eyes with PRR, associated VH occurred in 111 patients (80%). The median interval between the plaque application and the VH was 29 months (range, 1–139 months). Sixty-



**Figure 1.** A, Color fundus photograph from a 61-year-old patient with neovascularization elsewhere (NVE) temporal to the tumor scar 5 years after plaque radiotherapy for choroidal melanoma in the left eye. B, Fluorescein angiogram from the same patient demonstrating early hyperfluorescence corresponding to the NVE. C, Later frame of the angiogram in the same patient showing leakage in the area of the NVE. D, Color fundus photograph obtained 3 years after application of panretinal photocoagulation (PRP) laser showing that the NVE is completely resolved. E, Color fundus photograph from a 66-year-old patient with neovascularization at the disc (NVD) 2 years after plaque brachytherapy for a juxtapapillary choroidal melanoma. F, Fluorescein angiogram from the same patient showing hyperfluorescence at the disc on the early frames. G, Later frames of the angiogram of the same patient demonstrating leakage from the area of NVD. H, Color fundus photograph from the same patient showing resolution of the NVD 3 years after treatment with intravitreal bevacizumab injection and laser PRP.

seven patients with VH (49%) had associated retinal vein occlusion, and nonproliferative diabetic retinopathy was present in 8 patients (6%). The number of episodes of VH was 1 in 95 patients (69%), 2 in 14 patients (10%), and 3 in 2 patients (1%). Management of PRR included observation in 23 patients (17%), PRP in 96

patients (70%), and pars plana vitrectomy with endolaser photocoagulation in 29 (21%) patients. Enucleation was necessary in 15 patients (11%) because of neovascular glaucoma in 12 cases and tumor recurrence in 3 cases. The final outcome of the PRR in those patients who did not undergo enucleation was complete resolution

Table 2. Proliferative Radiation Retinopathy after Plaque Radiotherapy for Posterior Uveal Melanoma: Tumor Characteristics

| Features                          | Proliferative Radiation Retinopathy Group (n = 138), No. (%) | Control Group (n = 3703), No. (%) | P Value* |
|-----------------------------------|--------------------------------------------------------------|-----------------------------------|----------|
| Tumor tissue                      |                                                              |                                   |          |
| Choroid                           | 131 (95)                                                     | 3161 (85)                         |          |
| Ciliary body+choroid              | 7 (5)                                                        | 499 (13)                          | 0.036    |
| Iris+ciliary body+choroid         | 0 (0)                                                        | 43 (1)                            | 0.976    |
| Quadrant of tumor epicenter       |                                                              |                                   |          |
| Superior                          | 33 (24)                                                      | 973 (26)                          |          |
| Nasal                             | 30 (22)                                                      | 691 (19)                          | 0.306    |
| Inferior                          | 31 (22)                                                      | 688 (19)                          | 0.324    |
| Temporal                          | 31 (22)                                                      | 1092 (29)                         | 0.328    |
| Macular ( $\leq 3$ mm to fovea)   | 13 (9)                                                       | 259 (7)                           | 0.086    |
| Distance to optic nerve (mm)      |                                                              |                                   |          |
| Median (mean, range)              | 1.9 (2.8, 0–15)                                              | 4 (4.9, 0–20)                     | <0.001   |
| Distance to foveola (mm)          |                                                              |                                   |          |
| Median (mean, range)              | 2.5 (3, 0–12)                                                | 3.5 (4.7, 0–22)                   | <0.001   |
| Location of anterior tumor margin |                                                              |                                   |          |
| Macula to equator                 | 83 (60)                                                      | 1800 (49)                         |          |
| Equator to ora                    | 48 (35)                                                      | 984 (27)                          | 0.371    |
| Ciliary body                      | 7 (5)                                                        | 704 (19)                          | <0.001   |
| Iris                              | 0 (0)                                                        | 215 (6)                           | 0.944    |
| Largest base (mm)                 |                                                              |                                   | 0.757    |
| Median (mean, range)              | 10 (10, 5–18)                                                | 11 (10.8, 2–22)                   |          |
| Thickness (mm)                    |                                                              |                                   | 0.518    |
| Median (mean, range)              | 4 (4.8, 1.7–10.9)                                            | 4.2 (4.9, 0.25–15.5)              |          |
| Shape                             |                                                              |                                   |          |
| Dome                              | 96 (70)                                                      | 2909 (79)                         |          |
| Mushroom                          | 31 (22)                                                      | 541 (15)                          | 0.001    |
| Diffuse                           | 11 (8)                                                       | 253 (7)                           | 0.292    |
| Associated features               |                                                              |                                   |          |
| Bruch's membrane rupture          | 40 (29)                                                      | 541 (15)                          | <0.001   |
| Retinal invasion                  | 13 (9)                                                       | 286 (8)                           | 0.253    |
| Extraocular extension             | 1 (<1)                                                       | 56 (2)                            | 0.440    |
| Presence of subretinal fluid      |                                                              |                                   | 0.001    |
| No                                | 18 (13)                                                      | 924 (25)                          |          |
| Yes                               | 120 (87)                                                     | 2779 (75)                         |          |

\*Cox regression analysis.

in 78 patients (63%), partial resolution in 13 patients (11%), stable findings in 29 patients (24%), and progressive neovascularization in 3 patients (2%). Of the 96 patients who underwent PRP, 63 (66%) had complete resolution of the PRR. The occurrence of other radiation-associated side effects is summarized in Table 4. The median follow-up between the plaque brachytherapy application and the last recorded visit was 59 months (range, 4–173 months). Final visual acuity was 20/20 to 20/40 in 15 patients (11%), 20/50 to 20/100 in 13 patients (9%), and 20/200 or worse in 110 patients (80%).

By univariate analysis, the features predictive of development of PRR ( $P < 0.05$ ) included young age at presentation, diabetes mellitus, Hispanic race, hypertension, shorter tumor distance to the optic nerve and to the foveola, break in Bruch's membrane, choroidal location of the tumor, the presence of subretinal fluid, notched plaque, associated TTT treatment, radiation rate to the tumor apex and to the tumor base, and radiation dose to the optic nerve and to the foveola (Table 5). By multivariable analysis, young age, diabetes mellitus, and shorter tumor distance to the optic disc were the only factors predictive of development PRR (Table 5). By Kaplan-Meier analysis, the development of PRR was found in 1.7% of patients (95% CI, 1.2–2.2) by 2 years, 5.8% of patients (95% CI, 4.8–6.9) by 5 years, and 7% of patients (95%

CI, 5.7–8.2) by 10 and 15 years (Fig 2). In Figure 3, Kaplan-Meier analysis for time to onset of PRR according to the presence or absence of diabetes is shown.

## Discussion

Radiation retinopathy is a known cause of posterior segment neovascularization.<sup>7–28</sup> It has a predilection to affect the macula and presents with capillary occlusion, intraretinal microvascular abnormalities (telangiectasia, microaneurysms), vascular leakage (hard exudation, retinal edema), and neovascularization of the disc or retina (Fig 1).<sup>7–28</sup> In 1981, Irvine et al<sup>13</sup> described an animal model of PRR. They delivered 2000 to 8000 Rads of radiation to the eyes of 10 primates; PRR developed in 2 (20%) after 12 to 24 months of follow-up. Histopathologic study of these 2 eyes showed endothelial damage with ischemic vaso-occlusive disease very similar to that seen in diabetic retinopathy.<sup>13</sup>

The incidence of PRR after radiation therapy for the treatment of uveal melanoma ranges from 3% to 25%,



Table 3. Proliferative Radiation Retinopathy after Plaque Radiotherapy for Posterior Uveal Melanoma: Plaque Radiotherapy Parameters

| Features                             | Proliferative Radiation Retinopathy Group (n = 138), No. (%) | Control Group (n = 3703), No. (%) | P Value* |
|--------------------------------------|--------------------------------------------------------------|-----------------------------------|----------|
| Isotope                              |                                                              |                                   |          |
| Iodine 125                           | 133 (96)                                                     | 3126 (84)                         |          |
| Ruthenium 106                        | 0 (0)                                                        | 71 (2)                            | 0.938    |
| Cobalt 60                            | 3 (2)                                                        | 390 (11)                          | <0.001   |
| Iridium 192                          | 2 (1)                                                        | 116 (3)                           | 0.050    |
| Plaque shape                         |                                                              |                                   | <0.001   |
| Round                                | 78 (57)                                                      | 2846 (77)                         |          |
| Notched                              | 60 (43)                                                      | 857 (23)                          |          |
| Plaque size (mm)                     |                                                              |                                   |          |
| Median (mean, range)                 | 15 (16, 10–22)                                               | 15 (16, 8–25)                     | 0.011    |
| Hours of radiation                   |                                                              |                                   |          |
| Median (mean, range)                 | 97 (101, 58–188)                                             | 97 (115, 23–511)                  | <0.001   |
| Radiation to apex (Gy)               |                                                              |                                   |          |
| Median (mean, range)                 | 86.50 (89.59, 65–130)                                        | 83.05 (88.30, 12–300)             | 0.287    |
| Radiation rate at apex (cGy/hour)    |                                                              |                                   |          |
| Median (mean, range)                 | 87 (92, 52–155)                                              | 85 (85, 6–1089)                   | 0.025    |
| Radiation to base (Gy)               |                                                              |                                   |          |
| Median (mean, range)                 | 300 (295.39, 117–546)                                        | 300 (293.38, 25.50–815)           | 0.157    |
| Radiation rate at base (cGy/hour)    |                                                              |                                   |          |
| Median (mean, range)                 | 295 (297, 127–571)                                           | 264 (273, 14–3388)                | 0.011    |
| Radiation to disc (Gy)               |                                                              |                                   |          |
| Median (mean, range)                 | 59.12 (71.64, 7.53–312.79)                                   | 34.06 (50.16, 9–162.30)           | 0.002    |
| Radiation rate to disc (cGy/hour)    |                                                              |                                   |          |
| Median (mean, range)                 | 58 (73, 8–282)                                               | 36 (55, 0.5–5503)                 | 0.447    |
| Radiation to foveola (Gy)            |                                                              |                                   |          |
| Median (mean, range)                 | 60.80 (95.68, 8.18–471.84)                                   | 40.34 (66.77, 8–233.19)           | 0.007    |
| Radiation rate at foveola (cGy/hour) |                                                              |                                   |          |
| Median (mean, range)                 | 66 (97, 0–421)                                               | 42 (67, 1–528)                    | <0.001   |
| Radiation to lens (Gy)               |                                                              |                                   |          |
| Median (mean, range)                 | 16.28 (24.02, 14–93.31)                                      | 17.60 (27.47, 5–487.76)           | 0.461    |
| Radiation rate at lens (cGy/hour)    |                                                              |                                   |          |
| Median (mean, range)                 | 15 (19, 3–115)                                               | 16 (30, 0.24–3044)                | 0.030    |

cGy = centigray; Gy = gray.  
 \*Cox regression analysis.

according to different series.<sup>9–12</sup> Gragoudas et al<sup>12</sup> described the occurrence of PRR in 17 (3%) of 558 patients after proton beam irradiation of uveal melanoma. In that series, 11 patients had NVE, 6 patients had NVD, and 1 patient had a combination of both.<sup>12</sup> In the Collaborative Ocular Melanoma Study report number 30, Boldt et al<sup>7</sup> described a prevalence of NVD at 5 years of 5.2% in 650 patients who underwent plaque brachytherapy for medium uveal melanoma. In that report, NVE was not described and small and large tumors were not included. Gunduz et al<sup>11</sup> reported an incidence of PRR (NVE and NVD) of 3% after a median follow-up of 63 months in 630 patients who underwent plaque radiotherapy for uveal melanoma with macular involvement. The same authors reported that the proportion of patients with PRR in a series of 1300 eyes that had undergone plaque radiotherapy for uveal melanoma was 8% at 5 years of follow-up.<sup>9</sup> Shields et al<sup>10</sup> found that PRR developed in 25% of patients treated with plaque for large uveal melanoma ( $\geq 8$  mm in thickness) after 5 years of follow-up. In the present study, using Kaplan-Meier estimates, PRR developed in 6% of a total of 3841 patients at 5 years of follow-up (Fig 2).

Both patient and tumor baseline factors previously have been found to contribute to the development of PRR. Sagoo et al<sup>29</sup> reported that PRR developed in 25% of 37 patients with circumpapillary choroidal melanoma after plaque radiotherapy with a median follow-up of 46 months. The same authors described an incidence of PRR of 22% in a series of 141 patients with juxtapapillary choroidal melanoma after a mean follow-up of 56 months.<sup>30</sup> Boldt et al<sup>7</sup> found that shorter tumor distance to the optic nerve and the foveola as well as higher radiation dose to those 2 structures were prognostic factors for more prevalent and severe posterior pole abnormalities in 650 patients with medium choroidal melanoma treated with plaque radiotherapy. Tumor proximity to the optic disc and to the foveola were found to be risk factors for the development of PRR in this series, as well as the presence of a break in Bruch's membrane, choroidal location of the tumor (vs. ciliary body location), and the presence of subretinal fluid.

Diabetes previously was linked to the development of PRR. Gunduz et al<sup>9</sup> found that the most important risk factors for the development of PRR after plaque radiotherapy (multivariate analysis) were diabetes, Ir<sup>192</sup> isotope, and

Table 4. Proliferative Radiation Retinopathy after Plaque Radiotherapy for Posterior Uveal Melanoma: Associated Side Effects

| Features                                                 | Proliferative Radiation Retinopathy Group (n = 138), No. (%) | Control Group (n = 3703), No. (%) |
|----------------------------------------------------------|--------------------------------------------------------------|-----------------------------------|
| NPRR                                                     | 103 (75)                                                     | 1217 (33)                         |
| Interval from plaque to NPRR (mos)                       |                                                              |                                   |
| Median (mean, range)                                     | 17 (20, 0–88)                                                | 21 (29, 0–211)                    |
| Radiation maculopathy                                    | 86 (62)                                                      | 870 (23)                          |
| Interval from plaque to radiation maculopathy (mos)      |                                                              |                                   |
| Median (mean, range)                                     | 22 (27, 2–122)                                               | 22 (28, 1–151)                    |
| Radiation papillopathy                                   | 53 (38)                                                      | 456 (12)                          |
| Interval from plaque to radiation papillopathy (mos)     |                                                              |                                   |
| Median (mean, range)                                     | 22 (28, 5–129)                                               | 26 (33, 1–197)                    |
| Radiation cataract                                       | 83 (60)                                                      | 1435 (39)                         |
| Interval from plaque to radiation cataract (mos)         |                                                              |                                   |
| Median (mean, range)                                     | 31 (37, 2–138)                                               | 28 (36, 1–194)                    |
| Postradiation glaucoma                                   | 33 (24)                                                      | 218 (6)                           |
| Interval from plaque to glaucoma (mos)                   |                                                              |                                   |
| Median (mean, range)                                     | 31 (36, 11–85)                                               | 30 (40, 1–302)                    |
| Radiation scleral necrosis                               | 1 (<1)                                                       | 53 (1)                            |
| Interval from plaque to radiation scleral necrosis (mos) |                                                              |                                   |
| Median (mean, range)                                     | 23 (23, 23–23)                                               | 21 (39, 1–544)                    |

NPRR = nonproliferative radiation retinopathy.

basal tumor diameter of more than 10 mm. Kinyoun et al<sup>8</sup> identified vitreous hemorrhage, optic neuropathy, and macular ischemia as the main risk factors for PRR in a small series of 14 patients, although these probably represent

associated events rather than causative factors. In that report, diabetes was not found to be a risk factor for PRR. Boldt et al<sup>7</sup> described diabetes as one of the main risk factors for development of alterations in fluorescein angiog-

Table 5. Factors Predicting Proliferative Radiation Retinopathy: Cox Regression Analysis

| Variable                                               | Proliferative Radiation Retinopathy Group (n = 138), No. (%) | Control Group (n = 3703), No. (%) | P Value | Hazard Ratio       | 95% Confidence Interval |
|--------------------------------------------------------|--------------------------------------------------------------|-----------------------------------|---------|--------------------|-------------------------|
| Univariate analysis                                    |                                                              |                                   |         |                    |                         |
| Mean age (yrs)                                         | 52.14                                                        | 58.79                             | <0.001  | 1.30*              | 1.46–1.15               |
| Race (Hispanic vs. white <sup>†</sup> )                | 4 (2.9)                                                      | 36 (1.0)                          | 0.001   | 5.74               | 2.12–15.54              |
| Diabetes (yes vs. no <sup>†</sup> )                    | 28 (20.3)                                                    | 328 (8.9)                         | <0.001  | 2.94               | 1.94–4.45               |
| Hypertension (yes vs. no <sup>†</sup> )                | 37 (27.8)                                                    | 845 (22.8)                        | 0.005   | 1.71               | 1.17–2.48               |
| Tumor location (choroid vs. ciliochroid <sup>†</sup> ) | 131 (94.9)                                                   | 3161 (85.4)                       | 0.036   | 2.25               | 1.05–4.82               |
| Mean distance to optic nerve (mm)                      | 2.77                                                         | 4.88                              | <0.001  | 1.23 <sup>‡</sup>  | 1.15–1.31               |
| Mean distance to foveola (mm)                          | 3.07                                                         | 4.67                              | <0.001  | 1.13 <sup>‡</sup>  | 1.07–1.19               |
| Bruch's membrane rupture (yes vs. no <sup>†</sup> )    | 40 (29.0)                                                    | 541 (14.6)                        | <0.001  | 2.65               | 1.83–3.84               |
| SRF (yes vs. no <sup>†</sup> )                         | 120 (87.0)                                                   | 2779 (75.0)                       | 0.002   | 2.38               | 1.45–3.91               |
| Plaque shape (notched vs. round <sup>†</sup> )         | 60 (43.5)                                                    | 838 (22.7)                        | <0.001  | 2.67               | 1.91–3.74               |
| Mean radiation rate apex                               | 91.5                                                         | 84.9                              | 0.025   | 1.03 <sup>§</sup>  | 1.003–1.05              |
| Mean radiation rate base                               | 297.44                                                       | 272.81                            | 0.011   | 1.01 <sup>§</sup>  | 1.002–1.02              |
| Mean radiation disc                                    | 7164.21                                                      | 5016.45                           | 0.002   | 1.02 <sup>  </sup> | 1.007–1.03              |
| Mean radiation foveola                                 | 9568.46                                                      | 6676.99                           | 0.007   | 1.01 <sup>  </sup> | 1.003–1.02              |
| Mean radiation rate foveola                            | 97.31                                                        | 67.68                             | <0.001  | 1.04 <sup>§</sup>  | 1.02–1.07               |
| Adjusted therapy (TTT vs. none <sup>†</sup> )          | 56 (40.6)                                                    | 929 (25.1)                        | <0.001  | 2.85               | 2.02–4.02               |
| Multivariable analysis                                 |                                                              |                                   |         |                    |                         |
| Mean age (yrs)                                         | —                                                            | —                                 | <0.001  | 1.44*              | 1.25–1.67               |
| Diabetes (yes vs. no <sup>†</sup> )                    | —                                                            | —                                 | <0.001  | 2.73               | 1.69–4.40               |
| Mean distance to disc (mm)                             | —                                                            | —                                 | 0.011   | 1.10 <sup>‡</sup>  | 1.04–1.17               |

SRF = subretinal fluid; TTT = transpupillary thermotherapy.

\*Per 10-year decrease.

†Reference variable.

‡Per 1-mm decrease.

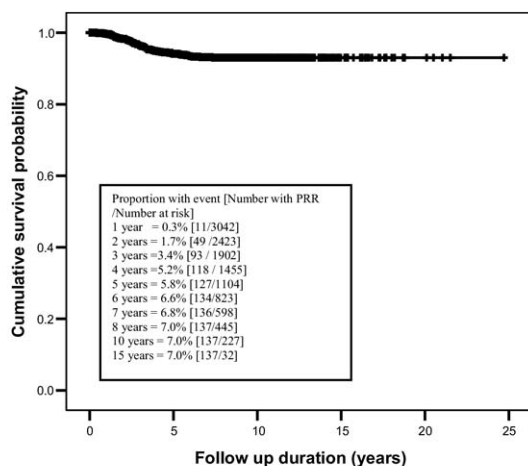
§Per 10-cGy/hour increase.

||Per 1000-cGy/hour increase.

raphy in 650 patients with medium choroidal melanoma after plaque brachytherapy. In the present study, diabetes was found to be a risk factor for the development of PRR on the multivariate analysis (Fig 3).

In this series, young patient age at presentation also was found to be a risk factor for the development of PRR in the multivariate model. In a series of 558 patients who underwent proton beam irradiation for the treatment of choroidal melanoma, Gragoudas et al<sup>12</sup> found that young patient age was a significant risk factor for the development of radiation maculopathy and papillopathy. Age also has been found to have an inverse relationship with the development of both proliferative and nonproliferative diabetic retinopathy. In a population-based cross-sectional study in Singapore, Wong et al<sup>31</sup> found a protective effect of older age in the development of any type of diabetic retinopathy (odds ratio, 0.73). Similar findings have been described previously by Klein et al.<sup>32–34</sup>

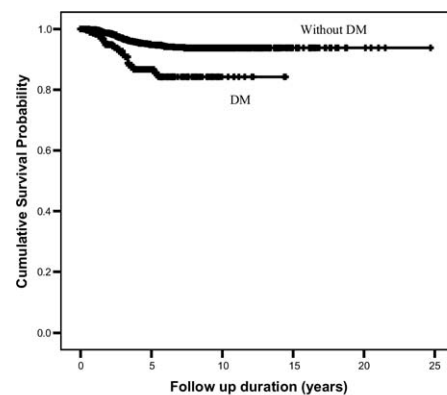
Panretinal photocoagulation caused complete resolution of the neovascularization in 66% of the cases in this study. Similarly, Kinyoun et al,<sup>8</sup> in a series of 14 patients with PRR after radiotherapy for head and neck tumor, found panretinal photocoagulation to be effective in the treatment of new blood vessels, with a resolution of the NVE and NVD in 92% of the cases. Management of PRR has similarities to the treatment of proliferative diabetic retinopathy.<sup>5–7,28</sup> Therefore, anti-vascular endothelial growth factor medications also should be considered in the treatment of PRR, because their potential role in the management of this condition may be significant. The high number of patients found to have branch retinal vein occlusion in the PRR group (49%) may be related to the application of associated TTT, because the later was found to be a risk factor for neovascularization in the univariate analysis.



Summary of outcomes using Kaplan-Meier estimates

| Outcomes                            | % at 2 years<br>[95% CI] | % at 5 years<br>[95% CI] | % at 10 years<br>[95% CI] | % at 15 years<br>[95% CI] |
|-------------------------------------|--------------------------|--------------------------|---------------------------|---------------------------|
| Proliferative Radiation Retinopathy | 1.7<br>[1.2 – 2.2]       | 5.8<br>[4.8–6.9]         | 7.0<br>[5.7 – 8.2]        | 7.0<br>[5.7 – 8.2]        |

Figure 2. Kaplan-Meier analysis showing time to onset of proliferative radiation retinopathy (PRR). CI = confidence interval.



| PRR – Proportion with event [Number with PRR / Number at risk] |                   |                   |                   |                   |                    |                   |                   |                   |                   |
|----------------------------------------------------------------|-------------------|-------------------|-------------------|-------------------|--------------------|-------------------|-------------------|-------------------|-------------------|
| Characteristic                                                 | 1 year            | 2 years           | 3 years           | 4 years           | 5 years            | 6 years           | 7 years           | 10 years          | 15 years          |
| With DM                                                        | 1.3%<br>[4/268]   | 5.1%<br>[13/196]  | 8.5%<br>[19/147]  | 13.3%<br>[26/111] | 13.3%<br>[26/81]   | 15.7%<br>[28/54]  | 15.7%<br>[28/37]  | 15.7%<br>[28/9]   | -                 |
| Without DM                                                     | 0.24%<br>[7/2774] | 1.4%<br>[36/2227] | 3.1%<br>[70/1759] | 4.4%<br>[92/1344] | 5.2%<br>[101/1023] | 5.7%<br>[106/769] | 6.0%<br>[108/561] | 6.2%<br>[109/218] | 6.2%<br>[109/218] |

Figure 3. Kaplan-Meier analysis for time to onset of proliferative radiation retinopathy (PRR) according to the presence or absence of diabetes. DM = diabetes mellitus.

Limitations of this study include the use of retrospective data collection, the sometimes difficult evaluation of the fundus resulting from the presence of hazy media (in cases with associated vitreous hemorrhage or cataract), and the loss to follow-up in a number of patients. These weaknesses are minimized by the high number of patients included in the study and by an appropriate statistical analysis.

In summary, PRR developed in 7% at 10 and 15 years using Kaplan-Meier analysis in 3841 patients after plaque radiotherapy for uveal melanoma. The mean time from plaque radiotherapy to onset of PRR was 32 months. Multivariate analysis showed that young patient age, diabetes, and short tumor distance to the optic disc were related to the occurrence of PRR. Panretinal photocoagulation was effective in causing resolution of the neovascularization in two thirds of the cases. Besides radiation exposure, baseline patient and tumor factors play a role in the development of PRR.

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Ocular Oncology Service, Wills Eye Institute, Thomas Jefferson University, Philadelphia, Pennsylvania.

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### Correspondence:

Carol L. Shields, MD, Ocular Oncology Service, Suite 1440, Wills Eye Institute, 840 Walnut Street, Philadelphia, PA 19107. E-mail: [carol.shields@shieldsoncology.com](mailto:carol.shields@shieldsoncology.com).